Sleep Paralysis Risk Prediction Module

This part does time window prediction, not real time muscle detection.

Core Idea

Use historical patient sleep and attack data to:

Predict which 15 minute time slots of next night have high probability of sleep paralysis.

Then send those time windows to ESP32 for vibration intervention.

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2. Objective

1. Predict high risk time periods of sleep paralysis using patient historical data
2. Use XGBoost time series model for risk prediction
3. Send predicted high risk slots to ESP32 device
4. Trigger controlled vibration to reduce or prevent paralysis episode

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3. Proposed Solution

The system consists of:

Software Module

- Collect sleep and attack history data
- Perform feature engineering
- Create lag and temporal features
- Train XGBoost classifier
- Predict probability per 15 minute slot for next night
- Select high risk slots using threshold

Hardware Module

- ESP32 based wearable device
- Real time clock
- Vibration motor
- Trigger vibration for 2 minutes every 15 minutes during high risk window

Reference

Sleep and Breathing 2025, Detection of sleep apnea using only inertial measurement unit signals from Apple Watch. LINK

1. What We Should Copy From This Paper

Not the topic, but the structure.

They followed this scientific flow:

1. Clear problem statement
2. Clinical gold standard reference
3. Feature extraction methodology
4. Machine learning comparison
5. Training and testing split
6. Feature importance analysis
7. Correlation with clinical severity
8. Discussion and limitations

We should follow the same structure for sleep paralysis.

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2. How Our Project Becomes "Similar but Unique"

Sleep Apnea Paper
Detect respiratory events per 30 second epoch

Our Project
Predict high risk time slot per 15 minute window

Both are:

Time segmented
Machine learning based
Wearable integrated
Clinically structured

Difference is:

They detect physiological event
We predict temporal risk window, That is acceptable and research valid.

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3. How To Adapt Their Structure To Our Project

Section 1: Introduction

Explain:

Sleep paralysis prevalence
Lack of preventive systems
Need for personalized risk prediction

Section 2: Methodology

Instead of IMU signal processing, write:

Data Collection
Sleep logs
Attack logs
Lifestyle variables

Time segmentation
15 minute slots

Summary Comparison

Sleep Apnea Study Uses
AHI
ODI
REI

WE USES:

Dataset A: Raw Feature Style (Calculate first manually)

Use this:

#	Feature	Your PDF Description
1	Sleep_Start	Sleep timing information
2	Wake_Time	Sleep timing information
3	Sleep_Duration	Sleep quantity
4	Sleep_Efficiency	Sleep quality proxy
5	Stress	Psychological stress level
6	HRV_ms	Physiological stress marker
7	Attack	Attack occurrence indicator
8	Attack_Time	Attack timing
9	REM_Vulnerability_Index	Attacks linked to REM stage
10	Mean_Attack_Interval	Recurrence rhythm
11	Attack_Time_STD	Time stability of attacks
12	Anxiety_Score	Pre-sleep anxiety — spikes sharply on attack nights
13	Caffeine_hrs_before_bed	REM suppression → rebound, 0 = no caffeine
14	Alcohol	REM suppression flag, 0/1
15	Sleep_Debt_3d	REM rebound pressure — Target(8h) minus avg last 3 nights
16	days_since_last_attack	Attack clustering, 999 = no prior history
17	SPHI	Sleep Paralysis Historical Index = attacks÷30

This is already strong behavioral modeling.

Dataset B : REM_timeline

REM_start
REM_End
Stage

(Calculate through apple watch XML report and add data entries in sheet)

https://www.johngoldin.com/blog/apple-health-export/2023-02-sleep-export/

sample image:

startDate	endDate	value
2023-01-11 02:40:03	2023-01-11 03:12:03	HKCategoryValueSleepAnalysisAsleepCore
2023-01-11 02:40:03	2023-01-11 03:25:03	HKCategoryValueSleepAnalysisInBed
2023-01-11 03:12:03	2023-01-11 03:25:03	HKCategoryValueSleepAnalysisAsleepREM
2023-01-11 03:25:03	2023-01-11 03:26:03	HKCategoryValueSleepAnalysisAwake
2023-01-11 03:26:03	2023-01-11 03:28:33	HKCategoryValueSleepAnalysisAsleepCore
2023-01-11 03:26:03	2023-01-11 06:05:03	HKCategoryValueSleepAnalysisInBed
2023-01-11 03:28:33	2023-01-11 03:32:03	HKCategoryValueSleepAnalysisAsleepREM

Dataset C: Final Dataset (Automatically Calculated using python code (Feature Engineering)

#	Feature	Description
1	**hour_sin**	Cyclical time encoding — part 1
2	**hour_cos**	Cyclical time encoding — part 2
3	**minutes_since_sleep_onset**	REM cycle position indicator — strongest timing feature
4	**REM_slot**	Biological REM alignment — direct mechanism feature
5	**Label**	Attack occurrence indicator — target variable

OTHER FEATURE ARE AS SAME AS ABOVE DATASET

SPHI replaces attack_last_7_days
SPTI replaces attack_frequency_this_slot_last_30_days
SPVI replaces sleep_irregularity_score

Below are equivalent structured indices for our system.

1. SPHI

Sleep Paralysis Historical Index

Definition
Total number of sleep paralysis episodes in last 30 days divided by total nights observed

Formula

SPHI = Total Attacks in 30 Days / 30

Purpose
Measures baseline vulnerability of patient

Comparable to
AHI in sleep apnea

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2. SPTI

Sleep Paralysis Time Index

Definition
Frequency of attack occurrence in a specific time slot

Formula

SPTI(slot) = Attacks in that slot / Total nights

Purpose
Measures circadian risk concentration

This is one of our strongest features.

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3. SPVI

Sleep Pattern Variability Index

Definition
Standard deviation of sleep start time over last 7 days

Formula

SPVI = Std Dev of Sleep Start Time

Purpose
Measures sleep irregularity

High value means unstable sleep schedule.

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4. REM Vulnerability Index

Definition
Attack frequency occurring after 90 minutes from sleep onset

Purpose : If most attacks occur during REM → vulnerability is high.
How strongly sleep paralysis attacks are linked to REM stage.

HRV (Heart Rate Variability)

What It Represents

HRV measures variation between heartbeats.
Low HRV usually indicates stress or sympathetic nervous system activation.

Why It Can Help

Sleep paralysis is often linked to:

• Stress
• Anxiety
• Irregular autonomic regulation

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5. Attack Recurrence Interval

Definition
Average number of days between two attacks

Purpose
Captures clustering behavior.

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6. SP Risk Score

We can combine multiple indices into one weighted score.

Example

SP Risk Score =
0.3 × SPHI

- 0.25 × SPTI
- 0.2 × SPDI
- 0.15 × SPVI
- 0.1 × Stress Score

This gives one interpretable clinical metric.

7 .Cyclical Encoding hour_sin, hour_cos (Divide attacks like clusters)

Sleep paralysis is time dependent.

It often happens during:

Early morning
REM cycles

If we use raw hour: Model cannot understand circular closeness. Its linear.

EX. 11 PM and 12 AM are only 1 hour apart. (23-0 =23, Predication wrong)

If we use cyclical encoding:

Formula:

hour_decimal = hour + minutes/60

hour_sin = sin(2π × hour_decimal / 24)
hour_cos = cos(2π × hour_decimal / 24)

This maps time onto a circle.

Model understands that 11:45 PM and 12:00 AM are close.

Example 1: Slot at 2:00 AM

2:00 AM
hour = 2

hour_sin ≈ 0.52
hour_cos ≈ 0.85

Label = 1 (attack happened)

Example 2: Slot at 2:30 AM

2:30 AM
hour = 2.5

hour_sin ≈ 0.61
hour_cos ≈ 0.79

Label = 1

Example 3: Slot at 2:45 AM

2:45 AM
hour ≈ 2.75

hour_sin ≈ 0.68
hour_cos ≈ 0.73

Label = 1

Notice something:

All these attack slots have similar hour_sin and hour_cos values.

So XGBoost sees pattern like:

If hour_sin around 0.5 to 0.7
and hour_cos around 0.7 to 0.9
→ high probability of attack

Now compare with 10:00 PM

10:00 PM = 22

hour_sin ≈ -0.5
hour_cos ≈ 0.87

Very different pattern.

XGBoost creates splits like:

If hour_sin > 0.45
and hour_cos > 0.7
→ Increase attack probability

That is how it detects time pattern.

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1. Directly Useful Data of SMARTWATCH for Our Project :

These are very important for prediction.

A. Sleep Start Time

When user falls asleep.

Why useful
Helps calculate sleep duration and REM cycles.

B. Wake Time

When user wakes up.

Used to compute total sleep duration.

C. Sleep Duration

Total hours slept.

Used to compute SPDI.

D. Sleep Stage Data

Light sleep
Deep sleep
REM sleep

Why important
Sleep paralysis happens during REM.
We can create REM vulnerability index using this.

E. REM Timeline

We have a REM segment: (Downloaded from smartwatch XML sheet)

It uses:

- Heart rate patterns
- Movement from accelerometer
- Respiratory rate

So REM segments are automatically detected by Apple's algorithm.

23:10 to 23:30 → Stage = REM

REM_SLOT_INDEX :

Now we divide the night into 15 minute slots.

Assume sleep started at 22:00.

Slots will be:

23:00–23:15 : 1
23:15–23:30 : 1
and other will be 0

If a slot is REM = 1

Probability increases.

That is direct biological alignment.

F. Heart Rate Variability

High stress loours HRV.

We can use HRV as stress indicator instead of manual stress input.

G. Bedtime Consistency : Smartwatch tracks daily sleep timing automatically.
Can compute SPVI automatically.

Section 3: Machine Learning

Do same approach:

Train
Test split

Compare:

Logistic Regression
Random Forest
XGBoost

Select best model

Section 4: Evaluation Metrics

Use:

AUC
Accuracy
Precision
Recall
F1 score

And also:

Correlation between predicted risk score and actual attack frequency

That mirrors their AHI correlation.

Section 5: Feature Importance

Like their FSI importance graph,
We show which feature most impacts prediction:

Maybe SPTI
Days since last attack
Sleep irregularity

This strengthens thesis quality.

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5. Why This Is a Smart Choice

Because:

1. It is wearable based

2. It uses machine learning
3. It has strong evaluation
4. It is published in 2025
5. It validates ML plus wearable approach

It gives credibility to our approach.

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1. Expected Output from XGBoost of Sleep Paralysis detection:

For each 15 minute slot of next night:

Model outputs:

Risk probability between 0 and 1

Example output:

Slot	Time	Risk Probability
1	10:00 PM	0.12
8	12:00 AM	0.68
9	12:15 AM	0.72
10	12:30 AM	0.65
20	3:00 AM	0.18

This is raw model output.

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2. Decision Logic

We must convert probability into actionable window.

Step 1

Choose threshold

Example
Threshold = 0.70

If probability > 0.70
Mark as High Risk

From above example:
12:15 AM

These become high risk window.

Step 2(IF MULIPLE SLOTS)

Merge consecutive slots

If multiple slots are continuous:

Combine into one window.

Example:

12:00 AM to 12:45 AM

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3. Final Expected Output Format

Our final system output should be:

High Risk Window 1:

Start: 00:00

End: 00:45

Risk Score: High

High Risk Window 2:

Start: 03:15

End: 03:30

Risk Score: Moderate

This is what we send to ESP32.

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4. How It Interacts With ESP32

There are 3 possible architectures.

Option 1: Bluetooth Direct Connection

Flow:

Laptop or mobile app
→ Sends time window
→ Via Bluetooth
→ ESP32 receives
→ Stores in memory

ESP32 logic:

If current_time within high_risk_window:
Activate vibration motor
2 minutes vibration every 15 minutes

Option 2: Cloud Based

Flow:

Model runs on cloud
→ High risk slots saved in database
→ ESP32 connects via WiFi
→ Fetches today's schedule

Better for real deployment.

Option 3: Mobile App Controller

Flow:

Model runs in mobile app
→ App sends schedule to ESP32
→ ESP32 acts as slave device

Best for prototype.

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5. ESP32 Internal Logic

Pseudo logic inside ESP32:

Get current time from RTC

Check if current time between start and end

If yes:

Turn vibration ON for 2 minutes

Wait 13 minutes

Repeat until window ends

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6. Complete System Architecture

Evening:
Model predicts risk slots

Night:
ESP32 executes schedule

Morning:
User logs whether attack occurred

Data saved
Model retrained periodically

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7. What We Will Show in Final Year Demo

1. Prediction table
2. Selected risk window
3. ESP32 receiving schedule
4. Vibration motor activating at correct time

This clearly demonstrates:

Prediction
Communication
Intervention

Final Clear Definition of Output

Our system output is:

List of high risk time windows with probability score, transmitted to ESP32 for scheduled vibration intervention.

(60 days of data from even 1 to 2 patients is much better than 4 to 5 patients with only few days.)

1. What Does Our Model Actually Need?

Our model predicts time slot pattern.

That means it must learn:

• Personal sleep rhythm
• Personal attack timing
• Personal irregularity trend

Sleep paralysis is highly personal.

It is not like diabetes where population model works easily.

So our model benefits more from long time history than many different people with short data.

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2. Compare Two Scenarios

Scenario A
5 patients
10 days each

Total = 50 days

Problem
Too few attacks per person
Model cannot learn time pattern
High risk of overfitting

Scenario B
1 patient
60 days

Total = 60 days

Better because:

Model sees repeated patterns
Learns circadian cycle
Learns REM trend
Learns clustering behavior

This is much stronger for time based prediction.

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3. Ideal Research Setup

Best design for final year:

2 to 3 patients
Each with 60 days data

Absolute Minimum

At least 30-40 attack events.(2-3 per week.)

Anything below 10 is unstable.

This gives:

Personalization, Some variability, Enough time history.

SAMPLE DAILY_LOG (RAW DATASET OF 7 DAYS) :

Patient_ID	Date	Sleep_Start	Wake_Time	Sleep_Duration	Sleep_Efficiency	HRV_ms	Stress	Anxiety_Score	Caffeine_hrs_before_bed	Alcohol	Sleep_Debt_3d	Attack	Attack_Time	days_since_last_attack	Mean_Attack_Interval	Attack_Time_STD	SPHI	REM_Vulnerability_Index
P1	2026-01-01	22:12	05:57	7.76	0.974	33	6	6	0	1	0.24	0		999	0	0	0	0.78
P1	2026-01-02	22:24	06:04	7.66	0.941	46	5	5	2.4	0	0.29	1	03:29	999	0	0	0	0.78
P1	2026-01-03	21:47	05:30	7.72	0.967	40	6	6	5.6	0	0.29	1	01:10	1	1	0	0.0333	0.78
P1	2026-01-04	22:20	05:57	7.62	0.936	49	5	6	4.2	0	0.33	0		1	1	69.31	0.0667	0.78
P1	2026-01-05	22:00	05:25	7.42	0.961	50	4	5	0	0	0.41	0		2	1	69.31	0.0667	0.78
P1	2026-01-06	22:05	06:09	8.06	0.985	41	6	5	0	0	0.3	1	03:01	3	1	69.31	0.0667	0.78
P1	2026-01-07	21:43	05:16	7.55	0.931	37	3	3	6.4	0	0.32	0		1	2	59.88	0.1	0.78
P1	2026-01-08	22:16	06:16	8	0.981	43	5	5	6.8	0	0.13	1	01:35	2	2	59.88	0.1	0.78
P1	2026-01-09	22:30	06:24	7.91	0.976	36	8	7	1.1	1	0.18	1	01:52	1	2	57.67	0.1333	0.78
P1	2026-01-10	22:20	06:55	8.59	0.972	33	6	6	7.4	0	-0.17	0		1	1.75	52.63	0.1667	0.78
P1	2026-01-11	22:20	05:57	7.62	0.957	39	5	5	0	0	-0.04	1	01:37	2	1.75	52.63	0.1667	0.78
P1	2026-01-12	21:54	05:35	7.69	0.956	44	6	6	4.6	0	0.03	1	02:49	1	1.8	49.93	0.2	0.78
P1	2026-01-13	21:48	05:26	7.63	0.963	36	6	5	0	0	0.35	1	02:41	1	1.67	48.48	0.2333	0.78
P1	2026-01-14	21:40	05:29	7.82	0.933	61	7	6	4.9	0	0.29	1	02:33	1	1.57	46.27	0.2667	0.78
P1	2026-01-15	21:32	04:56	7.4	0.978	36	7	6	5.8	0	0.38	0		1	1.5	43.94	0.3	0.78
P1	2026-01-16	22:13	06:15	8.04	0.95	45	7	6	0	0	0.25	1	01:34	2	1.5	43.94	0.3	0.78
P1	2026-01-17	22:15	06:00	7.75	0.938	35	6	6	0	0	0.27	0		1	1.56	43.79	0.3333	0.78
P1	2026-01-18	21:30	05:13	7.7	0.984	40	6	7	6.9	0	0.17	0		2	1.56	43.79	0.3333	0.78
P1	2026-01-19	21:39	05:38	7.99	0.935	51	6	6	5.3	0	0.19	0		3	1.56	43.79	0.3333	0.78
P1	2026-01-20	21:31	05:32	8.02	0.944	34	8	6	0	0	0.1	0		4	1.56	43.79	0.3333	0.78
P1	2026-01-21	21:39	05:30	7.84	0.952	34	5	5	0	0	0.05	0		5	1.56	43.79	0.3333	0.78
P1	2026-01-22	22:08	06:45	8.62	0.96	42	7	6	5.1	0	-0.16	0		6	1.56	43.79	0.3333	0.78
P1	2026-01-23	21:57	05:42	7.76	0.988	47	5	4	0	0	-0.08	1	01:15	7	1.56	43.79	0.3333	0.78
P1	2026-01-24	21:32	05:45	8.22	0.945	33	4	5	7.7	0	-0.2	1	02:36	1	2.1	45.12	0.3667	0.78
P1	2026-01-25	22:12	07:10	8.96	0.978	52	7	7	5.3	0	-0.31	1	01:33	1	2	43.86	0.4	0.78
P1	2026-01-26	21:38	05:18	7.66	0.989	43	6	6	2.2	0	-0.28	1	02:38	1	1.92	43.34	0.4333	0.78
P1	2026-01-27	22:57	07:19	8.35	0.968	25	6	6	6.3	1	-0.32	1	02:12	1	1.85	42.47	0.4667	0.78
P1	2026-01-28	22:43	06:19	7.6	0.933	41	7	6	4.7	0	0.13	0		1	1.79	41.03	0.5	0.78
P1	2026-01-29	22:51	06:43	7.87	0.956	32	6	6	3.8	1	0.06	1	03:52	2	1.79	41.03	0.5	0.78
P1	2026-01-30	21:38	05:25	7.78	0.959	31	6	5	0	0	0.25	1	00:52	1	1.8	46.76	0.5333	0.78
P2	2026-01-01	22:21	06:15	7.9	0.962	50	7	7	0	0	0.1	0		999	0	0	0	0.72
P2	2026-01-02	23:35	07:37	8.03	0.971	37	6	6	0	0	0.03	0		999	0	0	0	0.72
P2	2026-01-03	23:14	06:58	7.73	0.943	38	5	5	0	0	0.11	0		999	0	0	0	0.72
P2	2026-01-04	23:47	07:26	7.65	0.95	44	3	3	0	1	0.19	0		999	0	0	0	0.72
P2	2026-01-05	23:40	07:35	7.91	0.955	48	8	8	0	1	0.23	1	06:12	999	0	0	0	0.72
P2	2026-01-06	22:53	07:04	8.17	0.975	42	6	5	0	0	0.09	0		1	1	0	0.0333	0.72
P2	2026-01-07	23:28	07:27	7.98	0.987	44	8	7	4.8	0	-0.02	0		2	2	0	0.0333	0.72
P2	2026-01-08	22:24	06:27	8.06	0.935	49	4	5	0	0	-0.07	0		3	3	0	0.0333	0.72
P2	2026-01-09	22:35	06:24	7.82	0.939	49	7	6	0	0	0.05	1	01:51	4	4	0	0.0333	0.72
P2	2026-01-10	23:06	07:16	8.17	0.983	46	6	5	0	0	-0.01	0		1	4	130.15	0.0667	0.72
P2	2026-01-11	21:50	05:39	7.81	0.972	39	4	3	5.5	0	0.07	0		2	4	130.15	0.0667	0.72

ADDITIONAL FEATURES (REM_TIMELINE) :

Patient_ID	Date	REM_Start	REM_End	Stage
P1	2026-01-01	23:29	23:44	REM
P1	2026-01-01	01:06	01:27	REM
P1	2026-01-01	02:44	03:09	REM
P1	2026-01-01	04:10	04:43	REM
P1	2026-01-02	23:57	00:08	REM
P1	2026-01-02	01:30	01:51	REM
P1	2026-01-02	03:10	03:37	REM
P1	2026-01-02	04:35	05:05	REM
P1	2026-01-03	00:37	00:50	REM
P1	2026-01-03	02:08	02:29	REM
P1	2026-01-03	03:47	04:14	REM
P1	2026-01-03	05:10	05:42	REM
P1	2026-01-04	01:15	01:28	REM
P1	2026-01-04	02:43	03:03	REM
P1	2026-01-04	04:24	04:53	REM
P1	2026-01-04	05:46	06:19	REM
P1	2026-01-05	00:41	00:55	REM
P1	2026-01-05	02:04	02:25	REM
P1	2026-01-05	03:37	04:04	REM
P1	2026-01-05	05:00	05:34	REM
P1	2026-01-06	00:56	01:12	REM
P1	2026-01-06	02:29	02:53	REM
P1	2026-01-06	03:59	04:33	REM
P1	2026-01-07	23:51	00:05	REM

TRAINED USING PYTHON FILE, SLOT_DATASET FOR XGBoost train. :

Patient_ID	Date	Slot_Index	hour_sin	hour_cos	minutes_since_sleep_onset	REM_slot	REM_Vulnerability_Index	days_since_last_attack	SPHI	rolling_attack_mean_7	SPTI	Mean_Attack_Interval	Attack_Time_STD	Sleep_Duration	Sleep_Efficiency	Sleep_Debt_3d	SPVI	HRV_ms	Stress	Anxiety_Score	Caffeine_hrs_before_bed	Alcohol	Label	Split
P1	2026-01-05	17	0.743145	0.669131	255	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	18	0.785317	0.619094	270	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	19	0.824126	0.566406	285	1	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	20	0.859406	0.511293	300	1	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	21	0.891007	0.45399	315	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	22	0.918791	0.394744	330	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	23	0.942641	0.333807	345	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	24	0.962455	0.27144	360	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	25	0.978148	0.207912	375	1	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	26	0.989651	0.143493	390	1	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	27	0.996917	0.078459	405	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-05	28	0.999914	0.01309	420	0	0.8	999	0	0	0	0	0	7.38	0.954	0.5	38.02	32	8	8	0	0	0	train
P1	2026-01-06	0	-0.182236	0.983255	0	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	1	-0.117537	0.993068	15	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	2	-0.052336	0.99863	30	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	3	0.01309	0.999914	45	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	4	0.078459	0.996917	60	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	5	0.143493	0.989651	75	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	6	0.207912	0.978148	90	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	7	0.27144	0.962455	105	1	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	8	0.333807	0.942641	120	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	9	0.394744	0.918791	135	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	10	0.45399	0.891007	150	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	11	0.511293	0.859406	165	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	12	0.566406	0.824126	180	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	13	0.619094	0.785317	195	1	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	1	train
P1	2026-01-06	14	0.669131	0.743145	210	1	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	15	0.716302	0.69779	225	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	16	0.760406	0.649448	240	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	17	0.801254	0.598325	255	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	18	0.838671	0.544639	270	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	19	0.872496	0.488621	285	1	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	20	0.902585	0.430511	300	1	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	21	0.92881	0.370557	315	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	22	0.951057	0.309017	330	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	23	0.969231	0.246153	345	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-06	24	0.983255	0.182236	360	0	0.8	999	0	0	0	0	0	6.31	0.932	1	34.25	28	8	10	4.9	0	0	train
P1	2026-01-07	0	-0.430511	0.902585	0	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	1	-0.370557	0.92881	15	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	2	-0.309017	0.951057	30	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	3	-0.246153	0.969231	45	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	4	-0.182236	0.983255	60	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	5	-0.117537	0.993068	75	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	6	-0.052336	0.99863	90	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	7	0.01309	0.999914	105	1	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	8	0.078459	0.996917	120	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	9	0.143493	0.989651	135	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	10	0.207912	0.978148	150	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	11	0.27144	0.962455	165	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	12	0.333807	0.942641	180	0	1	1	0.033333	0.142857	0	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train
P1	2026-01-07	13	0.394744	0.918791	195	1	1	1	0.033333	0.142857	0.166667	1	0	7.74	0.967	0.86	33.1	43	5	7	0	0	0	train

There are approx.. 669 entries

3 Patients, 7 days, 32 slots of 15 min in 8 hrs(if there is 7.2 hrs sleep then entries may change), 32*7 = 224 entries of 1 patient P1.
Similar for 3(P1,P2,P3) * 224 = 672. (~669)